

04.02 Fractional Exponents

Common problems and solution methods

Rule guide: recognize the task first

Recognize	Apply these rules
An exponent has a denominator.	Use the denominator as the root index.
An exponent has a numerator.	Raise the base to that power.
The problem mixes powers and roots.	Rewrite in the form that makes simplification easiest, then apply exponent rules.

Example 1: Rewrite a fractional exponent

Problem	$x^{3/2}$
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How to solve it

Read the denominator:

$$x^{3/2} = (x^{1/2})^3$$

Write as a radical:

$$= (\sqrt{x})^3$$

Example 2: Evaluate a fractional power

Problem	$16^{3/4}$
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How to solve it

Take the fourth root:

$$16^{1/4} = 2$$

Raise to the third power:

$$16^{3/4} = 2^3 = 8$$

Example 3: Simplify a product

Problem	$x^{2/3} x^{1/3}$
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How to solve it

Add exponents:

$$x^{2/3} x^{1/3} = x^{2/3 + 1/3}$$

Simplify:

$$= x$$